

Pre-CyberSecurity

PreCyberSecurity is a beginner-friendly course designed to build a solid foundation before diving into the world of cybersecurity. You'll learn essential skills in networking, Linux, Windows, and basic cybersecurity concepts

Shatha Alosaimi

■ Whoami

```
# id  
uid=0(root) gid=0(root)  
groups=0(root)
```

- Info: Shatha Alosaimi, 22 yo, penetration tester, junior bug hunter, cybersecurity content creator on X, and senior management information systems student
- Certs: eJPTv2, eCPPTv2, eWPTv2, eWPTXv3, CRTA
- Interested in: Business, Financial, Marketing, investment, and systems analysis and design
- Social media:
 - # X
 - # LinkedIn

You will learn

- Basics of networking (LAN, protocols, OSI model, TCP/IP etc)
- Introduction to lab environment (Kali Linux distro)
- Linux fundamentals (What is linux, command lines)
- Windows fundamentals (What is windows, system files, Desktop environment etc)
- A little bit of cybersecurity fundamentals
- So now what? What is my next step?
- How to choose your field in cybersecurity (Bonus)
- A roadmap for starting cybersecurity (Bonus)

Pre requirement

- Basics of computer (I prefer CompTIA A+ course)
- PC or laptop with 8GB memory
- Basically a good connection (:
- It's recommended that you are comfortable with English
- And pay attention carefully to me

CompTIA A+
course



Let's Get Started

TryHackMe

- This course will be explained using selected rooms from TryHackMe platform
- TryHackMe is a free online platform for learning cyber security, using hands-on exercises and labs, all through your browser!
- Some of THM rooms need a subscription
- It covers a wide range of topics, from basic concepts to advanced penetration testing, allowing learners to practice skills in real-world simulated environments.
- Let's take a look on THM website

[TryHackMe URL](#)

Netowking



Network fundamentals

What is Network?

- Imagine if I told you, "I have a good network of connections." The first thing that comes to mind is that my network consists of two or more people. That's exactly the idea behind a computer network.
- **A network is simply two or more devices (such as computers, printers, or servers) connected together to share data. The connection can be through cables or wirelessly (Wi-Fi). Networks can be small (like your home network) or extremely large (like corporate networks or even the Internet).**

What is Internet?

- **The Internet is the “network of networks.” It’s the massive infrastructure that connects millions of networks worldwide, allowing us to send emails, browse websites, communicate, and learn. Think of it as the global “highway” for data.**

Network fundamentals

To communicate and maintain order, devices must be both identifying and identifiable on a network. What use is it if you don't know whom you're talking to at the end of the day?

Devices on a network are very similar to humans in the fact that we have two ways of being identified:

- **Our Name (IP Address in network)**
- **Our Fingerprints (MAC Address in network)**

What is an IP Address?

- **An IP address is a unique number assigned to each device on a network so we know who is sending and receiving data. It's like your home address, but for your device.**
- **Example: 192.168.1.1**
- **There are two types: IPv4 and IPv6.**

Public IP Vs. Private IP

IPv4 VS IPv6

- **IPv4: Uses 32-bit addresses, written as four decimal numbers separated by dots (e.g., 192.168.1.1). It supports about 4.3 billion unique addresses.**
- **IPv6: Uses 128-bit addresses, written in hexadecimal and separated by colons (e.g., 2001:0db8:85a3::8a2e:0370:7334). It supports an almost unlimited number of unique addresses and is designed to replace IPv4.**

Public IP

- **Assigned by your Internet Service Provider (ISP).**
- **Used to identify your network on the Internet.**
- **Accessible from anywhere in the world (unless blocked by a firewall).**

Private IP:

- **Used inside local networks (e.g., home, office).**
- **Cannot be accessed directly from the Internet.**
- **Common ranges: 192.168.x.x, 10.x.x.x, 172.16.x.x – 172.31.x.x.**

Public IP Vs. Private IP

Important note:

- Any IP that start with (10 - 172 - 192) in its first octet then it refers to a private IP address

Examples:
192.168.1.77 (Private)
86.157.52.21 (Public)

Device Name	IP Address	IP Address Type
DESKTOP-KJE57FD	192.168.1.77	Private
DESKTOP-KJE57FD	86.157.52.21	Public
CMNatic-PC	192.168.1.74	Private
CMNatic-PC	86.157.52.21	Public

MAC Address

What is MAC Address?

- A MAC address is a unique identifier assigned to a network interface card (NIC) in every device. It's permanent and does not change (unlike an IP address, which can change). Switches use it to know which device is connected to which port.
- Example: **00:1A:2B:3C:4D:5E**

Vendor who build the
network interface
(in this instance, Intel)

Unique address of the
network interface

Practical exercise on THM



a4 : c3 : f0 : 85 : ac : 2d

ping and ICMP

what is ping?

- The ping tool is used to test connectivity between two devices on a network. When you ping, the device sends an ICMP message (a small “Are you there?” request) to another device, and that device replies if it’s reachable.
- If you get a reply, there’s a connection. If not, the device may be turned off or blocked by a firewall.

ping command:

- Pings can be performed against devices on a network, and websites as well, this tool can be easily used and comes installed on Operating Systems (OSs) such as Linux and Windows
- The syntax to do a simple ping is: **ping IP address or website URL**

```
cmnatic@CMNatic-THM-LPTOP:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=63 time=2.15 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=63 time=3.34 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=63 time=10.3 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=63 time=4.45 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.152/4.160/10.313/2.864 ms
cmnatic@CMNatic-THM-LPTOP:~$
```

Practical exercise on THM

Networks and its own categories

- Networks are divided into two categories: first, by geographical coverage, and second, by topology.
- there are four categories of networks by its geographical coverage

1- LAN

2- PAN

3- MAN

4- WAN

What is LAN?

- **LAN (Local Area Network) is a type of network based on geographical coverage. It connects devices within a limited area such as a building, office, or home. A LAN allows users to share resources like printers, files, and Internet access.**

A topology... ?

- **Topology is the design or physical layout of a network, basically, how devices are arranged and connected to each other.**

Network topologies

1. Star Topology

The most common type of network. All devices connect to a central device (switch or hub).

- **Advantages: Easy to expand, faults are isolated.**
- **Disadvantages: If the central device fails, the entire network stops.**

2. Ring Topology

Each device connects to the next in a closed loop. Data travels in one or both directions.

- **Advantages: Consistent performance under moderate load.**
- **Disadvantages: A single break in the loop can bring down the network (unless it's bidirectional).**

3. Bus Topology

All devices connect to a single main cable called the backbone.

- **Advantages: Easy to set up, low cost.**
- **Disadvantages: If the backbone fails, the whole network fails. Performance decreases with more devices.**

Look at the pictures of
them on THM

Network Devices

- We will cover two devices of networks for now, switch and router

What is switch?

- **Switches are dedicated devices within a network that are designed to aggregate multiple other devices such as computers, printers, or any other networking-capable device using ethernet. These various devices plug into a switch's port. Switches are usually found in larger networks such as businesses, schools, or similar-sized networks, where there are many devices to connect to the network. Switches can connect a large number of devices by having ports of 4, 8, 16, 24, 32, and 64 for devices to plug into.**
- **Switches are much more efficient than their lesser counterpart (hubs/repeaters). Switches keep track of what device is connected to which port. This way, when they receive a packet, instead of repeating that packet to every port like a hub would do, it just sends it to the intended target, thus reducing network traffic.**



Look at the pictures of them on THM

Network Devices

What is router?

- It's a router's job to connect networks and pass data between them. It does this by using routing (hence the name router!).
- Routing is the label given to the process of data travelling across networks. Routing involves creating a path between networks so that this data can be successfully delivered.
- Both Switches and Routers can be connected to one another



Practical exercise on THM

Subnetting

What is subnetting?

- **Subnetting is the term given to splitting up a network into smaller, miniature networks within itself**
- Take a business, for example; You will have different departments such as: Accounting - Finance - Human Resources

Example: Network 192.168.1.0/24 can be split into:

- 192.168.1.0/25
- 192.168.1.128/25

Subnetting provides a range of benefits, including:

- **Efficiency**
- **Security**
- **Full control**

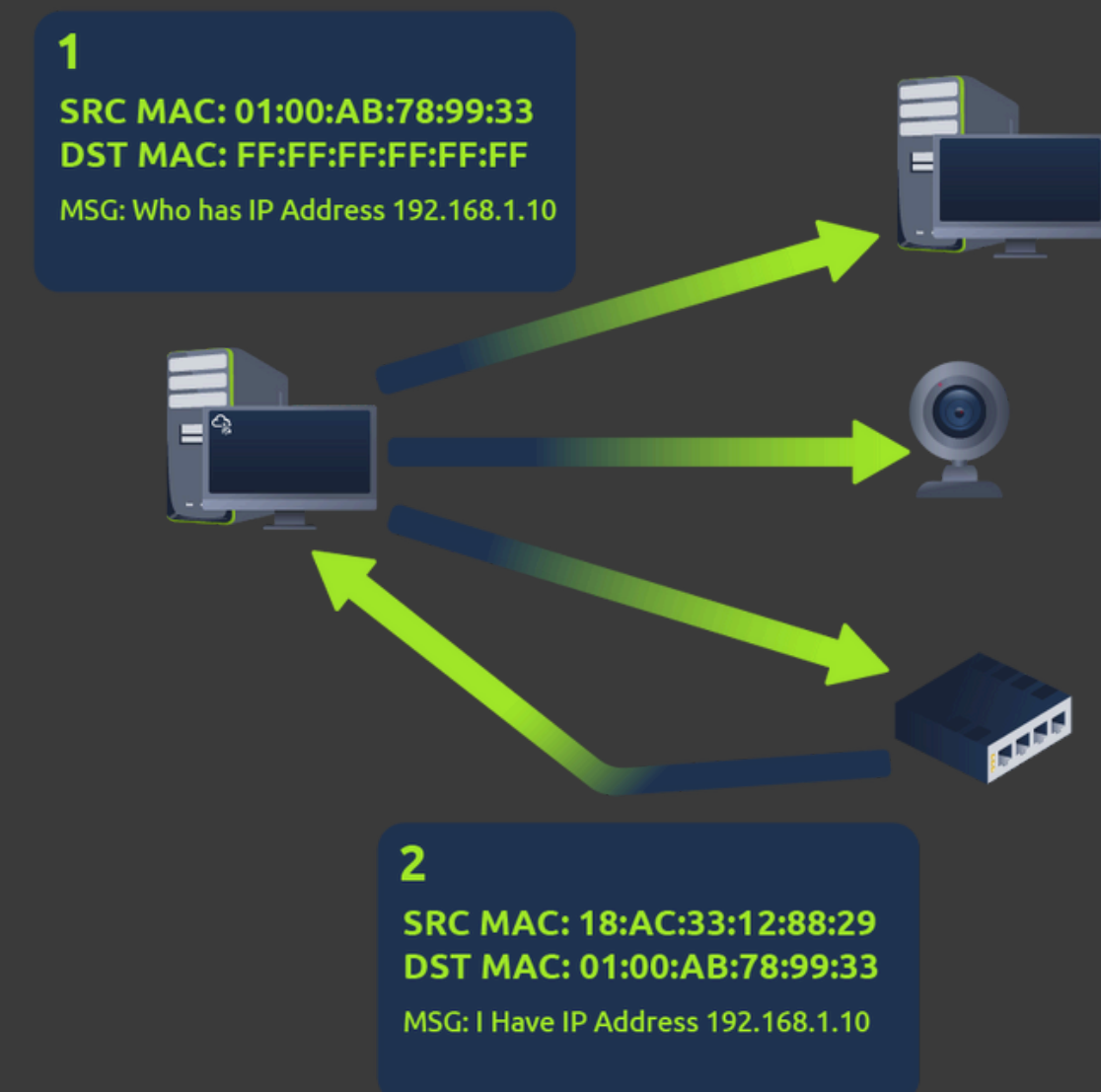
ARP

What is ARP?

- A protocol that works inside a local network (LAN) to map an IP address to a MAC address.

How it works:

- Your device broadcasts: "Who has IP address X.X.X.X?"
- The target device replies with its MAC address.
- The requesting device can now remember this mapping and store it in its ARP cache for future use

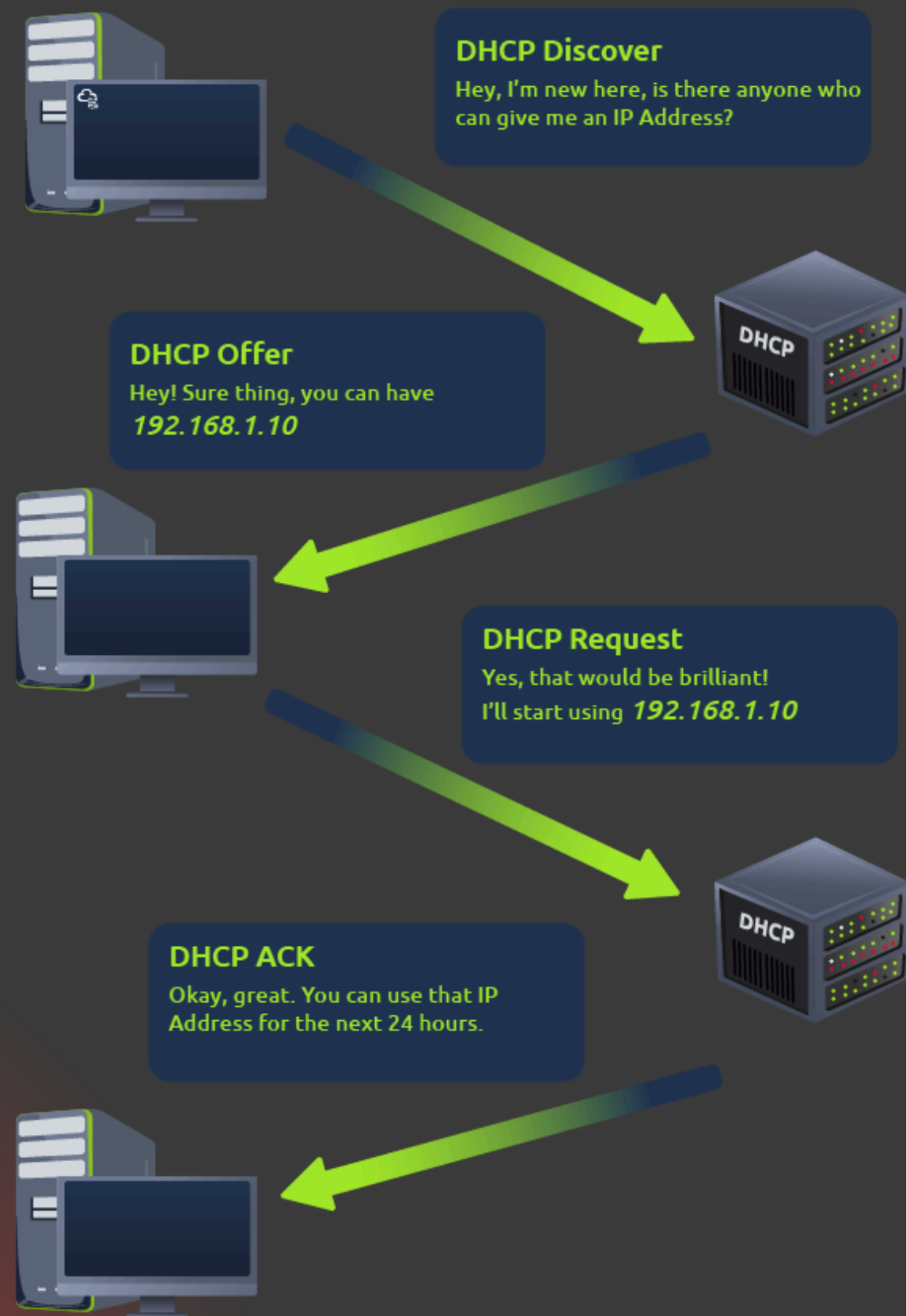


DHCP

What is DHCP?

- **IP addresses can be assigned either manually, by entering them physically into a device, or automatically and most commonly by using a DHCP (Dynamic Host Configuration Protocol) server. When a device connects to a network, if it has not already been manually assigned an IP address, it sends out a request (DHCP Discover) to see if any DHCP servers are on the network. The DHCP server then replies back with an IP address the device could use (DHCP Offer). The device then sends a reply confirming it wants the offered IP Address (DHCP Request), and then lastly, the DHCP server sends a reply acknowledging this has been completed, and the device can start using the IP Address (DHCP ACK).**

How DHCP works



OSI model

What is OSI model?

- **The OSI model (or Open Systems Interconnection Model) is an essential model used in networking. This critical model provides a framework dictating how all networked devices will send, receive and interpret data.**
- **One of the main benefits of the OSI model is that devices can have different functions and designs on a network while communicating with other devices. Data sent across a network that follows the uniformity of the OSI model can be understood by other devices.**

7. Application

6. Presentation

5. Session

4. Transport

3. Network

2. Data link

1. Physical

OSI model

Layer 7 – Application Layer

- **Closest to the end user.**
- **Provides network services to applications (e.g., web browsers, email clients).**
- **Examples: HTTP, HTTPS, FTP, SMTP, DNS.**

Layer 6 – Presentation Layer

- **Translates, encrypts, and compresses data.**
- **Ensures data is in a usable format for the Application layer.**
- **Examples: SSL/TLS encryption, JPEG, MP3.**

Layer 5 – Session Layer

- **Manages sessions between devices (start, maintain, and end connections).**
- **Examples: NetBIOS, RPC, PPTP.**

Layer 4 – Transport Layer

- **Responsible for reliable or unreliable delivery of data.**
- **Handles segmentation, flow control, and error correction.**
- **Examples: TCP, UDP.**

OSI model

Layer 3 – Network Layer

- **Routes data between devices across different networks.**
- **Uses logical addressing (IP addresses).**
- **Examples: IP, ICMP, OSPF.**
- **Data names as Packets**

Layer 2 – Data Link Layer

- **Transfers data between two devices in the same network.**
- **Uses physical addresses (MAC addresses).**
- **Examples: Ethernet, PPP, Switches.**
- **Data names as Frames**

Layer 1 – Physical Layer

- **The physical hardware and transmission of raw bits.**
- **Examples: Cables, Hubs, Wi-Fi signals, Fiber optics.**

OSI model summary

When data is sent from one device to another, it moves down the OSI layers on the sender's side and up the OSI layers on the receiver's side:

1. Application Layer (7) – The user's application (e.g., browser) creates the data to send.
2. Presentation Layer (6) – The data is encrypted, compressed, or translated into the proper format.
3. Session Layer (5) – A communication session is established with the receiving device.
4. Transport Layer (4) – The data is broken into segments, assigned port numbers, and prepared for delivery (TCP or UDP).
5. Network Layer (3) – Each segment gets an IP address for routing across networks.
6. Data Link Layer (2) – The IP packet is framed with the MAC address for delivery within the local network.
7. Physical Layer (1) – The frames are converted into electrical signals, light pulses, or radio waves and sent through the medium.

TCP and UDP

TCP (Transmission Control Protocol)

- Connection-oriented: Establishes a connection before sending data.
- Reliable: Ensures all data is delivered in the correct order.
- Slower due to error checking and acknowledgment.
- Use cases: Web browsing (HTTP/HTTPS), email (SMTP), file transfers (FTP).

Advantages:

- Reliable delivery of data.
- Ensures data order.
- Error detection and correction.

Disadvantages:

- Slower due to overhead.
- Requires more resources.n.
- Usage: email for example

TCP and UDP

UDP (User Datagram Protocol)

- Connectionless: Sends data without establishing a connection.
- Unreliable: No guarantee of delivery or order.
- Faster because it has less overhead.
- Use cases: Streaming, online gaming, VoIP, DNS queries.

Advantages:

- Faster transmission.
- Lower resource usage.

Disadvantages:

- No guarantee of delivery.
- No order control.
- No built-in error correction.
- usage: streaming

a very important note:

- TCP = "Reliable but slower"
- UDP = "Fast but no guarantees"

Packets and Frames

Packets and frames are small pieces of data that, when forming together, make a larger piece of information or message. However, they are two different things in the OSI model.

What is Packets?

- **A packet is a piece of data from Layer 3 (Network Layer) of the OSI model, containing information such as an IP header and payload**

What is Frames?

- **A frame, however, is used at Layer 2 (Data Link) of the OSI model, which, encapsulates the packet and adds additional information such as MAC addresses.**

Packets and Frames Headers

Some notable headers include:

- **Time to Live** > This field sets an expiry timer for the packet to not clog up your network if it never manages to reach a host or escape!
- **Checksum** > This field provides integrity checking for protocols such as TCP/IP. If any data is changed, this value will be different from what was expected and therefore corrupt.
- **Source Address** > The IP address of the device that the packet is being sent from so that data knows where to return to.
- **Destination Address** > The device's IP address the packet is being sent to so that data knows where to travel next.

TCP/IP

What is TCP/IP?

- **the foundational suite of communication protocols used to connect networks and devices on the internet and private networks**
- **Very similar to how the OSI model works, information is added to each layer of the TCP model as the piece of data (or packet) traverses it. As you may recall, this process is known as encapsulation - where the reverse of this process is decapsulation** (will talk about capsulation and decapsulation later)

The 4 Layers of TCP/IP:

1- Application: **User-facing apps (how data is displayed/used).**

- **Protocols: HTTP, FTP, DNS, SMTP**

2- Transport: **Ensures data delivery (TCP = reliable, UDP = fast).**

- **Protocols: TCP, UDP**

3- Internet: **Routes packets using IP addresses.**

- **Protocols: IP, ICMP, ARP**

4- Network Access: **Sends raw data over physical connections (cables, Wi-Fi).**

- **Protocols: Ethernet, Wi-Fi, PPP**

TCP/IP Vs. OSI model

Layer Number	ISO OSI Model	TCP/IP Model (RFC 1122)	Protocols
7	Application Layer	Application Layer	HTTP, HTTPS, FTP, POP3, SMTP, IMAP, Telnet, SSH,
6	Presentation Layer		
5	Session Layer		
4	Transport Layer		
3	Network Layer	Internet Layer	IP, ICMP, IPSec
2	Data Link Layer	Link Layer	Ethernet 802.3, WiFi 802.11
1	Physical Layer		

TCP Packet contents

- 1- Source Port > This value is the port opened by the sender to send the TCP packet from. This value is chosen randomly (out of the ports from 0-65535 that aren't already in use at the time).
- 2- Destination Port > This value is the port number that an application or service is running on the remote host (the one receiving data); for example, a webserver running on port 80. Unlike the source port, this value is not chosen at random.
- 3- Source IP > This is the IP address of the device that is sending the packet.
- 4- Destination IP > This is the IP address of the device that the packet is destined for.

TCP Packet contents

- 5- Sequence Number > When a connection occurs, the first piece of data transmitted is given a random number. We'll explain this more in-depth further on.
- 6- Acknowledgement Number > After a piece of data has been given a sequence number, the number for the next piece of data will have the sequence number + 1. We'll also explain this more in-depth further on.
- 7- Checksum > This value is what gives TCP integrity. A mathematical calculation is made where the output is remembered. When the receiving device performs the mathematical calculation, the data must be corrupt if the output is different from what was sent.
- 8- Data > This header is where the data, i.e. bytes of a file that is being transmitted, is stored.
- 9- Flag > This header determines how the packet should be handled by either device during the handshake process. Specific flags will determine specific behaviours, which is what we'll come on to explain below.

Three-way handshake

What is Three-way handshake?

- **The 3-way handshake is how two devices (like your computer and a web server) establish a reliable TCP connection before sending data. It ensures both sides are ready to communicate.**

Step-by-Step Process

1- SYN (Synchronize)

- **Client → Server:**
 - **Sends a SYN packet with a random sequence number (e.g., Seq=100).**
 - **Meaning: "Hey server, I want to connect. My starting sequence number is 100."**

2- SYN-ACK (Synchronize-Acknowledge)

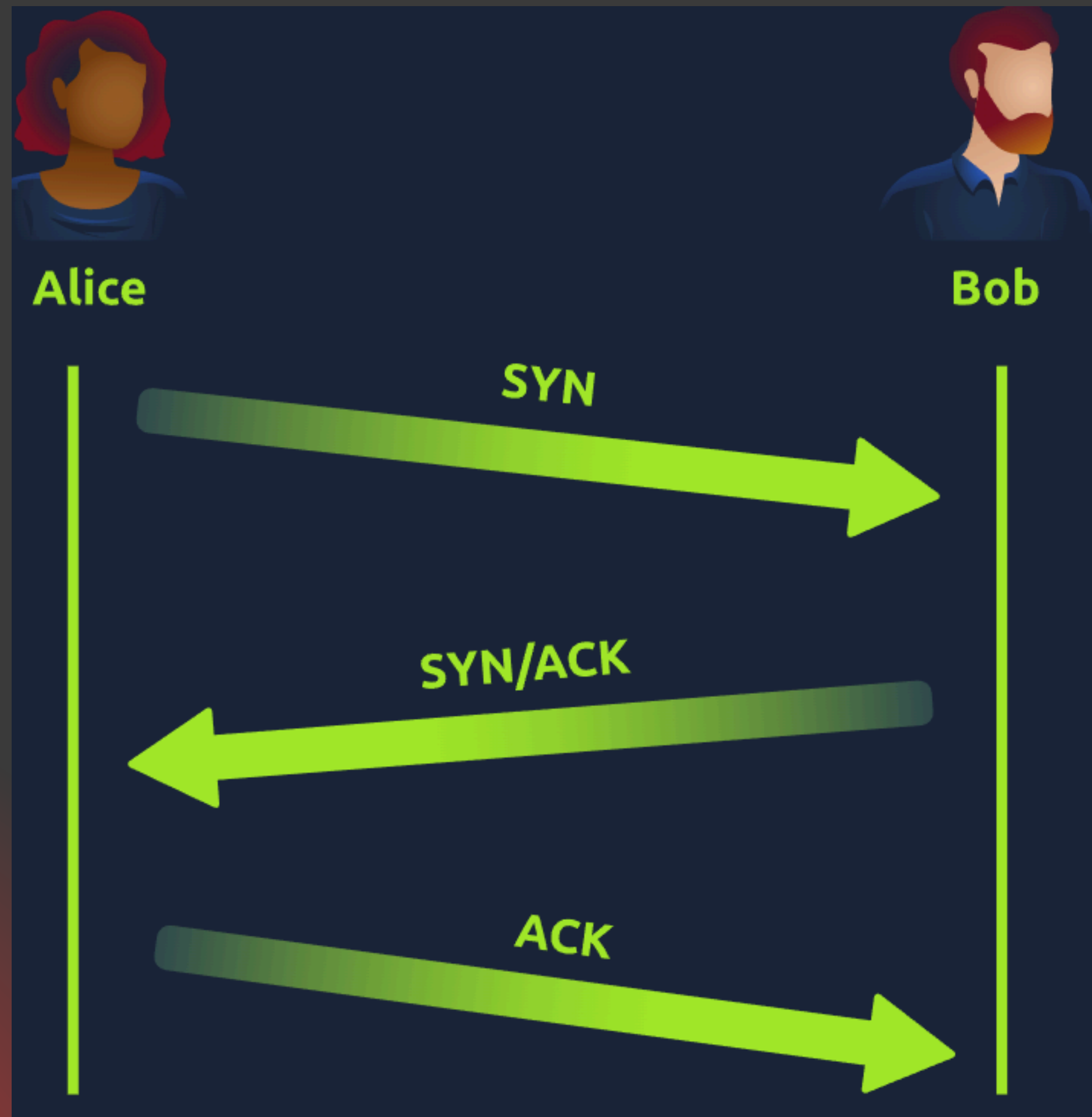
- **Server → Client:**
 - **Responds with SYN-ACK:**
 - **ACK = 101 (*"I got your SYN, next I expect Seq=101"*).**
 - **SYN with its own random sequence number (e.g., Seq=500).**
 - **Meaning: *"I acknowledge your request (ACK=101), and my starting sequence number is 500."***

3- ACK (Acknowledge)

- **Client → Server:**
 - **Sends ACK = 501 (*"I got your SYN, next I expect Seq=501"*).**
 - **Meaning: "Got it! Let's start sending data."**

✓ **Connection Established! Now data can flow reliably.**

How Three-way handshake works



Three-way handshake flags

1- SYN > A SYN message is the initial packet sent by a client during the handshake. This packet is used to initiate a connection and synchronise the two devices together (we'll explain this further later on).

2- SYN/ACK > This packet is sent by the receiving device (server) to acknowledge the synchronisation attempt from the client.

3- ACK > The acknowledgement packet can be used by either the client or server to acknowledge that a series of messages/packets have been successfully received.

4- DATA > Once a connection has been established, data (such as bytes of a file) is sent via the "DATA" message.

5- FIN > This packet is used to cleanly (properly) close the connection after it has been complete.

6- RST > This packet abruptly ends all communication. This is the last resort and indicates there was some problem during the process. For example, if the service or application is not working correctly, or the system has faults such as low resources.

TCP Closing a Connection



UDP/IP

UDP (User Datagram Protocol) is the "fast but reckless" cousin of TCP. It's used when speed matters more than reliability.

Unlike its brother TCP, UDP is a stateless protocol that doesn't require a constant connection between the two devices for data to be sent. For example, the Three-way handshake does not occur, nor is there any synchronisation between the two devices.

UDP Vs. TCP...

Feature	UDP	TCP (For Comparison)
Connection	✗ No handshake (connectionless).	✓ 3-way handshake (connection-oriented).
Reliability	✗ No retransmission of lost packets.	✓ Guaranteed delivery.
Order	✗ Packets may arrive out of order.	✓ In-order delivery.
Speed	⚡ Faster (less overhead).	⚠ Slower (due to error-checking).
Use Cases	Gaming, VoIP, live streams, DNS, DHCP.	Web browsing, emails, file transfers.

UDP packets content

1- Time to Live (TTL) > This field sets an expiry timer for the packet, so it doesn't clog up your network if it never manages to reach a host or escape!

2- Source Address > The IP address of the device that the packet is being sent from, so that data knows where to return to.

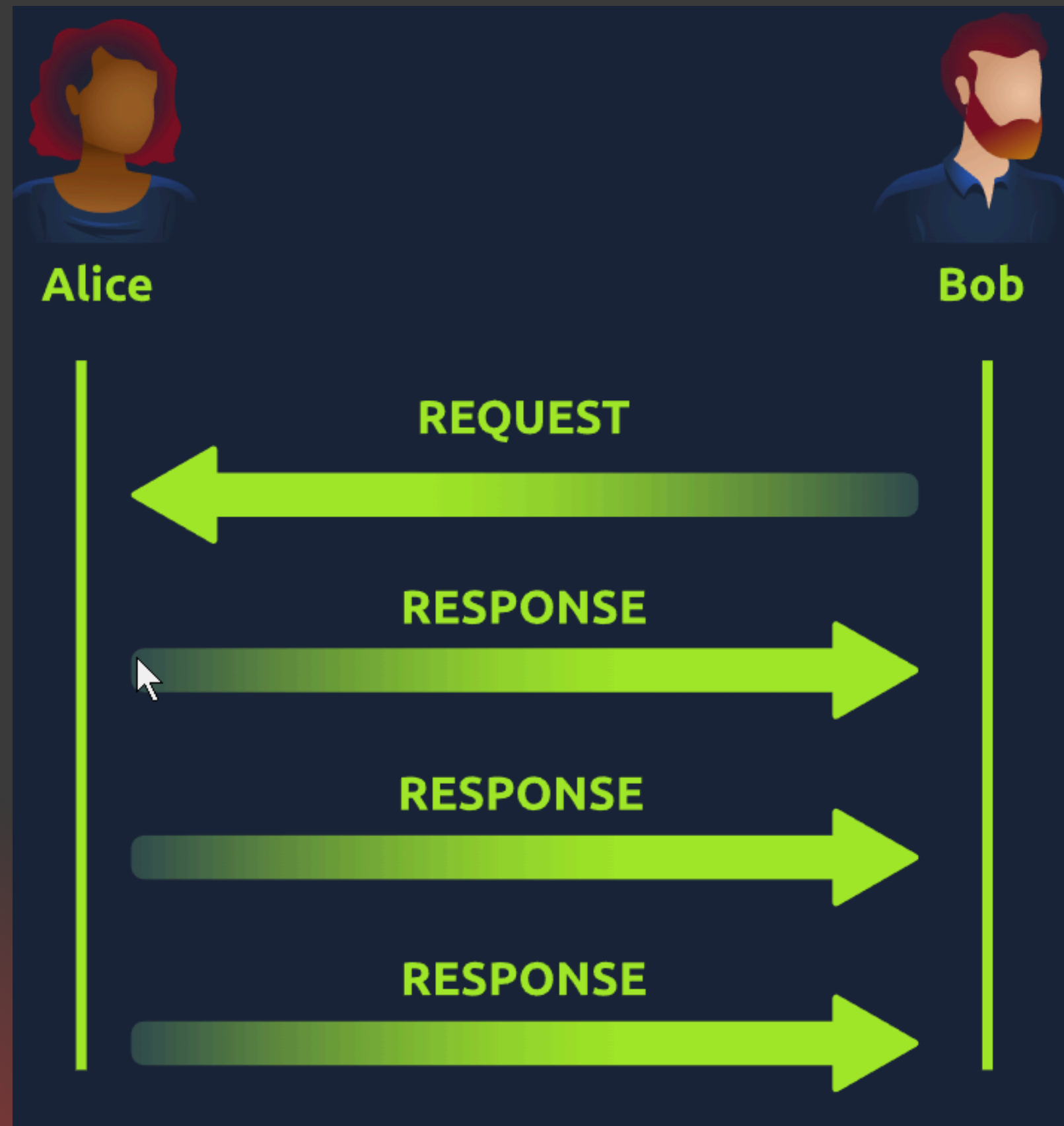
3- Destination Address > The device's IP address the packet is being sent to so that data knows where to travel next.

4- Source Port > This value is the port that is opened by the sender to send the UDP packet from. This value is randomly chosen (out of the ports from 0-65535 that aren't already in use at the time).

5- Destination Port > This value is the port number that an application or service is running on the remote host (the one receiving the data); for example, a webserver running on port 80. Unlike the source port, this value is not chosen at random.

6- Data > This header is where data, i.e. bytes of a file that is being transmitted, is stored.

How UDP works



Encapsulation vs. Decapsulation

1. Encapsulation (Adding Headers/Trailers)

- What? **Wrapping data with protocol-specific headers/trailers as it moves down the OSI/TCP/IP layers.**
- Why? **To prepare data for transmission across a network.**
- How? **Each layer adds its own header:**
 - *Application Layer: Original data (e.g., HTTP request).*
 - *Transport Layer (TCP/UDP): Adds source/dest ports + sequence numbers.*
 - *Network Layer (IP): Adds source/dest IP addresses.*
 - *Data Link Layer: Adds MAC addresses + FCS (error-checking trailer).*
 - *Physical Layer: Converts to bits (0s/1s) for cables/Wi-Fi.*

Encapsulation vs. Decapsulation

2. Decapsulation (Removing Headers/Trailers)

- What? **Stripping headers/trailers as data moves up the OSI/TCP/IP layers.**
- Why? **To deliver the original data to the correct application.**
- How? **Each layer removes its own header:**
 - *Physical Layer: Converts bits to frames.*
 - *Data Link Layer: Checks MAC → removes header/trailer.*
 - *Network Layer: Checks IP → removes IP header.*
 - *Transport Layer: Checks ports → removes TCP/UDP header.*
 - *Application Layer: Delivers raw data (e.g., webpage).*

Ports 101

What is Netowk Ports?

- **A port is a virtual "door" on a device (like your PC or server) that allows you to communicate with specific service**
- **Identified by numbers (0 to 65535).**

Most common ports that you must know at this moment:

A table of the 1024 common ports listed

Protocol	Port Number	Description
File Transfer Protocol (<u>FTP</u>)	21	This protocol is used by a file-sharing application built on a client-server model, meaning you can download files from a central location.
Secure Shell (<u>SSH</u>)	22	This protocol is used to securely login to systems via a text-based interface for management.
HyperText Transfer Protocol (<u>HTTP</u>)	80	This protocol powers the World Wide Web (WWW)! Your browser uses this to download text, images and videos of web pages.
HyperText Transfer Protocol Secure (<u>HTTPS</u>)	443	This protocol does the exact same as above; however, securely using encryption.
Server Message Block (<u>SMB</u>)	445	This protocol is similar to the File Transfer Protocol (<u>FTP</u>); however, as well as files, <u>SMB</u> allows you to share devices like printers.
Remote Desktop Protocol (<u>RDP</u>)	3389	This protocol is a secure means of logging in to a system using a visual desktop interface (as opposed to the text-based limitations of the <u>SSH</u> protocol).

Subnet Mask

What is Subnet Mask?

A subnet mask is a 32-bit number used in IP networks to divide an IP address into two parts:

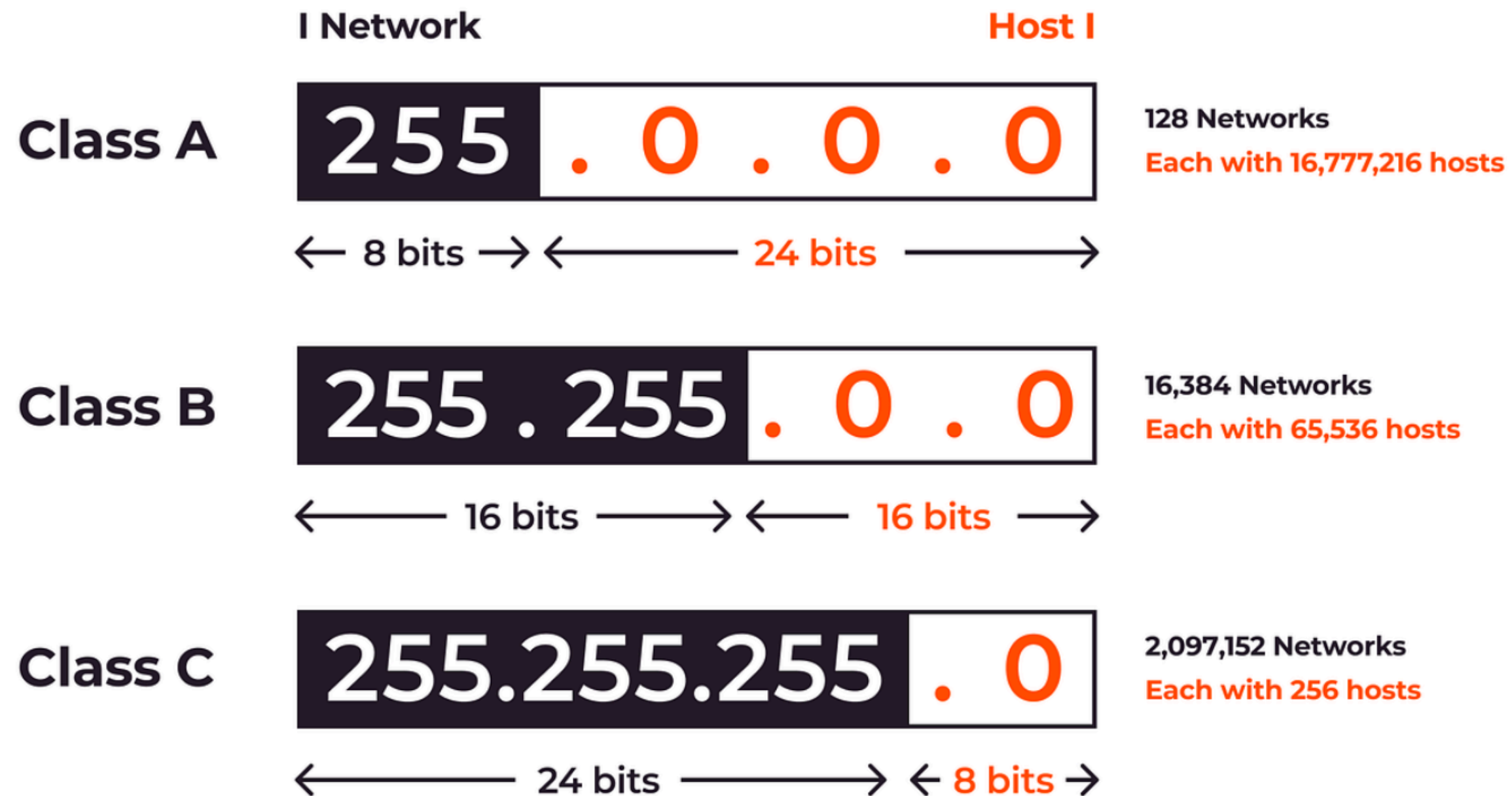
- **Network portion: identifies the specific network.**
- **Host portion: identifies the devices (hosts) within that network.**

For example, the subnet mask **255.255.255.0** (also written as /24) means that the first 24 bits of the IP address are fixed for the network, while the last 8 bits are available for hosts.

This allows IP addresses to range from 192.168.66.1 to 192.168.66.254, with 192.168.66.0 as the network address and 192.168.66.255 as the broadcast address.

Subnet Mask

IPv4 Classes and Subnet Masks



Looking Up Your Network Configuration

Windows:

you can check your IP address and subnet mask using the following commands:

ipconfig

Linux/UNIX:

you can check your IP address and subnet mask using the following commands:

- Linux/UNIX: ifconfig or ip address show (which can also be shortened to ip a s)

Take a look on TryHackMe example

NAT

Why NAT!!

IPv4 can technically support around 4 billion devices. However, with the explosion of devices (PCs, smartphones, IoT devices like cameras, TVs, and even washing machines), **the IPv4 space was not enough.**

To solve this problem, Network Address Translation (NAT) was introduced.

How NAT Works?

NAT allows **multiple private IP addresses inside a network to access the Internet using one public IP address.**

- Example: A company has 20 computers.
 - Without NAT → 20 public IP addresses would be needed.
 - With NAT → only 1 public IP address is used to provide Internet access to all 20 computers.

DNS

1# What is DNS?

DNS is what allows us to use human-readable names (like example.com) instead of memorizing IP addresses.

- Uses **UDP port 53** (default) and TCP port 53 (fallback).

DNS Records:

- **A record:** The A (Address) record maps a hostname to one or more IPv4 addresses. For example, you can set example.com to resolve to 172.17.2.172.
- **AAAA Record:** The AAAA record is similar to the A Record, but it is for IPv6. Remember that it is AAAA (quad-A), as AA and AAA would refer to a battery size; furthermore, AAA refers to Authentication, Authorization, and Accounting; neither falls under DNS.
- **CNAME Record:** The CNAME (Canonical Name) record maps a domain name to another domain name. For example, www.example.com can be mapped to example.com or even to example.org.
- **MX Record:** The MX (Mail Exchange) record specifies the mail server responsible for handling emails for a domain.
- If you want to **look up the IP address of a domain from the command line, you can use a tool such as nslookup**

WHOIS

What is WHOIS?

- WHOIS is a **public database that contains details about domain ownership, including registrant name, phone, email, and address.**
- Anyone can query WHOIS using tools like whois on Linux or online services.
- Privacy services can be used to hide personal information.

Example command: **whois google.com**

WHOIS online tool

```
(root@kali)-[~]
# whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
```

HTTP and HTTPS

What is HTTP and HTTPS

- **HTTP (Hypertext Transfer Protocol):** The foundation of data communication on the web. Runs on TCP port 80.
- **HTTPS (HTTP Secure):** Same as HTTP but encrypted using SSL/TLS. Runs on TCP port 443.
- **HTTPS ensures confidentiality, integrity, and authenticity of data.**
- **HTTPS doesn't mean that the website is secure and there is no risk, it only means that the data between you and web server are encrypted**

FTP

What is FTP

Unlike HTTP, which is designed to retrieve web pages, File Transfer Protocol (FTP) is designed to transfer files. As a result, FTP is very efficient for file transfer, and when all conditions are equal, it can achieve higher speeds than HTTP.

- Used to transfer files between client and server.
- **Runs on TCP port 21.**
- **Supports authentication with username and password.**
- **Not secure (credentials and data are sent in plaintext).**
- Secure alternative: SFTP (SSH File Transfer Protocol).

SMTP

What is SMTP

Simple Mail Transfer Protocol (SMTP) defines how a mail client talks with a mail server and how a mail server talks with another.

The analogy for the SMTP protocol is when you go to the local post office to send a package. You greet the employee, tell them where you want to send your package, and provide the sender's information before handing them the package. Depending on the country you are in, you might be asked to show your identity card. This process is not very different from an SMTP session.

- Used for sending emails between servers.
- **Runs on:**
 - Port 25 (traditional, but often blocked due to spam).
 - Port 587 (submission).
 - Port 465 (SMTP over SSL).

POP3

What is POP3

You've received an email and want to download it to your local mail client. The Post Office Protocol version 3 (POP3) is designed to allow the client to communicate with a mail server and retrieve email messages.

- **Used for retrieving emails from a mail server.**
- **Runs on TCP port 110.**
- **By default, downloads emails and removes them from the server (unless configured otherwise).**
- **Secure version: POP3S on port 995.**

IMAP

What is IMAP

what if you want to check your email from your office desktop computer and from your laptop or smartphone? In this scenario, you need a protocol that allows synchronization of messages instead of deleting a message after retrieving it. **One solution to maintaining a synchronized mailbox across multiple devices is Internet Message Access Protocol (IMAP).**

- **Runs on TCP port 143.**
 - **Keeps emails on the server, allowing access from multiple devices.**
 - **Secure version: IMAPS on port 993.**
-
- **SMTP → Send**
 - **POP3 → Receive (download & delete)**
 - **IMAP → Receive (sync & keep on server)**

TLS

Before Encryption

- In the past, attackers could use packet capturing tools (like Wireshark) to read all chats, emails, and passwords on the network.
- Since credentials were sent in cleartext, there was nothing users could do to protect themselves.
- **So TLS released to encrypts communication between applications over the Internet (mainly web traffic).**
- TLS works at the Transport Layer of the OSI model and ensures:
 - Confidentiality: No one can read the data.
 - Integrity: No one can modify the data.

Thanks to TLS, many protocols gained secure versions:

- **HTTP → HTTPS**
- **SMTP → SMTPS**
- **IMAP → IMAPS**

Telnet and SSH

What is Telnet

is a **client-server** application protocol that provides access to virtual terminals of remote systems on local area networks or the Internet.

- Telnet was useful for remote system administration **but sent all data in cleartext (easy for attackers to steal credentials).**
- Tatu Ylönen developed the Secure Shell (SSH) protocol and released SSH-1 in 1995

Benefits of SSH:

- **Secure Authentication: Supports password, public key, and 2FA.**
- **Confidentiality: End-to-end encryption protects against eavesdropping.**
- **Integrity: Prevents tampering with traffic.**

You would issue the command **ssh username@hostname** to connect to an SSH server. If the username is the same as your logged-in username, you only need **ssh hostname**. Then, you will be asked for a password; however, if public-key authentication is used, you will be logged in immediately.

VPN

What is VPN?

A Virtual Private Network (VPN) **is a technology that allows secure communication between devices over the internet as if they were physically connected to the same private network.** It is widely used by companies with multiple offices to share resources securely and cost-effectively.

- **Purpose:** Encrypts all network traffic between your device and a VPN server, hiding your real IP.
- **Scope:** System-wide encryption (not just one app, like TLS).
- **Think of it as:** A private tunnel through the Internet.

How VPN works?

- **A VPN client (e.g., in a remote office) connects to a VPN server (e.g., in the main branch).**
- **Data is encrypted and sent through a secure VPN tunnel (represented by blue lines in network diagrams).**
- **Once received, the traffic is decrypted (green lines) and routed internally.**

My next step

Your next step

You've learned a lot of concepts and networking fundamentals with me, but I recommend you go for a **Network+ or CCNA course, because the basics we covered still need deeper research and a stronger understanding of networking.** I suggest this two resources for you

CompTIA Network+ course:

<https://youtube.com/playlist?list=PLH-n8YK76vliulZoWvHL7AvtrDV7hR3He&si=fv2A-J2DqJRIRnJv>

CCNA course:

https://youtube.com/playlist?list=PLAqaqJU4wzYU_6ElzVoxHghQILgyY--yf&si=qKckTu9szB8bX8j9

Linux



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Linux

What is Linux?

Just like Windows, iOS, and Mac OS, **Linux is an open-source operating system -also a kernel-**. In fact, one of the most popular platforms on the planet, Android, is powered by the Linux operating system. An operating system is software that manages all of the hardware resources associated with your desktop or laptop.

Linux powers things such as:

- **Websites that you visit**
- **Car entertainment/control panels**
- **Point of Sale (PoS) systems such as checkout tills and registers in shops**
- **Critical infrastructures such as traffic light controllers or industrial sensors**

[More information](#)

Let's set our lab

Now we're going to learn how to download, use, and master Kali Linux. Here's what you'll need to do:

1. First, download either VMware or VirtualBox (whichever you prefer) from these links:

- [VMware Download](#)
- [VirtualBox Download](#)

2. Next, go to the official [Kali Linux website \(kali.org\)](https://kali.org) and download the latest available version.

3. Finally, install Kali Linux on your VMware or VirtualBox to set it up as a virtual machine.

LAB



Linux permissions

What is Linux permissions?

Linux **uses a permission system to control who can read, write, or execute files and directories.**

Types of Permissions:

1. **Read (r)** → view the contents of a file or list files in a directory.
2. **Write (w)** → modify a file or create/delete files in a directory.
3. **Execute (x)** → run a file (if it's a script or program) or access a directory.

Permission Groups:

- Owner (u) → the user who owns the file.
- Group (g) → users who are members of the file's group.
- Others (o) → everyone else.

The Differences Between Users & Groups

The idea is that Linux permissions are very flexible. Even if a file has a specific owner, you can give a group of users their own set of permissions for the same file. This doesn't change the owner's permissions.

For example, a web server user needs permission to read and write files for a website to work. But a hosting company can't just let customers use the web server's account (because that would be a big security risk). Instead, with Linux permissions, the company can let customers upload and manage their own files without giving them the same access as the web server user.

Linux common directories

- # **Root directory /** (top of the filesystem).
- # **/bin** → Essential binaries (basic commands like ls, cp, cat).
- # **/sbin** → System binaries (for admin tasks, like shutdown, mount).
- # **/etc** → Configuration files (system-wide settings).
- # **/home** → User home directories (e.g., /home/shatha).
- # **/root** → Home directory for the root user.
- # **/var** → Variable data (logs, caches, spool files).
- # **/tmp** → Temporary files (automatically cleaned on reboot).
- # **/usr** → User programs, libraries, and documentation.
- # **/opt** → Optional software (third-party apps).

Linux Terminal text editors

At first, we learned how to write text into files using the echo command together with pipe operators like > and >>. **While this works for simple cases, it quickly becomes inefficient if the file contains multiple lines or needs frequent editing. That's why we move on to using terminal text editors.**

Terminal text editors make writing and editing files much easier. There are several options available, but two of the most common are Nano and VIM. Nano is beginner-friendly and straightforward, while VIM is more advanced, offering powerful features but with a steeper learning curve.

nano

To open or edit a file using Nano, you simply type:

nano filename

Replace filename with the name of the file you want to create or edit. Once you press enter, Nano will launch. The interface allows you to immediately start typing, move around lines with the arrow keys, and create new lines with the Enter key.

When you run this command, Nano opens the file called myfile. You can type text directly into it and make changes on the spot. At the bottom of the editor, you'll see a list of shortcuts that let you save, exit, search, or perform other actions.

Nano Features

Nano comes with a handful of simple yet useful features. You can search for text, copy and paste lines, or jump directly to a specific line number. You can also check your current line number at any time. These features are accessed using the Ctrl key (shown as ^ in the interface). For example, pressing Ctrl + X exits Nano, while Ctrl + O saves your changes.

vim

VIM is a much more advanced text editor compared to Nano. Although it takes longer to get used to, it offers significant advantages. It is highly customizable, allowing you to modify shortcuts and behavior. It also supports syntax highlighting, which makes it especially useful for developers who write or maintain code. Another benefit is that VIM is available on all terminals, while Nano might not always be installed.

In short, Nano is simple, intuitive, and great for beginners who want to get things done quickly. VIM, on the other hand, requires more time to learn but provides powerful features that make it a favorite among experienced users. If you're interested in going deeper

Downloading a files using wget

One of the core features of computing is the ability to transfer files. Whether you want to download a program, a script, or even a picture, Linux provides different ways to handle this. In this lesson, we'll focus on three methods: **downloading files with wget, transferring files with scp, and serving files using Python's built-in web server.**

The wget command **lets us download files from the web over HTTP**, almost as if we were accessing them through a browser. All we need to do is provide the web address of the file. For example, if we want to download a file called myfile.txt, the command would look like this:

wget https://assets.tryhackme.com/additional/linux-fundamentals/part3/myfile.txt

This retrieves the file directly into our current directory.

Transferring Files with SCP

Another common situation is transferring files between two systems. The command scp (secure copy) **is designed for this purpose. It works like the regular cp command, but uses SSH for authentication and encryption.**

With scp, you can copy files from your local system to a remote machine, or from a remote machine to your local system.

For example, let's say we want to copy a file called important.txt from our local machine to a remote server with IP 192.168.1.30, under the user ubuntu. We also want to save it as transferred.txt on the remote system. The command would be:

scp important.txt ubuntu@192.168.1.30:/home/ubuntu/transferred.txt

192.168.1.30 > The IP address of the remote system

ubuntu > User on the remote system

important.txt > Name of the file on the local system

transferred.txt > Name that we wish to store the file as on the remote system

Copying Back with SCP

The reverse is also possible: downloading a file from a remote system to our local one. Suppose the remote system has a file called documents.txt in /home/ubuntu/, and we want to copy it back to our local machine as notes.txt. The command would look like this:

```
scp ubuntu@192.168.1.30:/home/ubuntu/documents.txt notes.txt
```

192.168.1.30 > IP address of the remote system

ubuntu > User on the remote system

documents.txt > Name of the file on the remote system

notes.txt > Name that we wish to store the file as on our system

Serving Files with Python Web Server

Sometimes, instead of pushing or pulling files directly, you may want to temporarily serve them from your machine so another system can download them. Python makes this easy with its built-in HTTPServer module.

By running the following command inside the directory containing your files, you instantly start a lightweight web server:

```
python3 -m http.server
```

This will serve files on port 8000 by default

Once the server is running, **you can download files from it using wget**. For example, if the server's IP is MACHINE_IP and you want to download myfile, the command would be:

```
wget http://MACHINE_IP:8000/myfile
```

It's important to run this command in a new terminal, because the terminal running the Python web server will stay occupied until you stop the server.

Linux Processes

What is process?

Processes are the programs that are running on your machine. They are managed by the kernel, where **each process will have an ID associated with it, also known as its PID.** The PID increments for the order in which the process starts. I.e. the 60th process will have a PID of 60.

Viewing processes:

- Use **ps** to view processes in the current session.
- **ps aux** shows processes of all users and system processes.
- Use **top** for real-time process monitoring (auto-refresh every 10s).

Managing Processes

- Processes can be terminated with signals using kill <PID>.
- For example: kill 1337

Common signals:

- SIGTERM – terminate, allow cleanup.
- SIGKILL – force kill, no cleanup.
- SIGSTOP – suspend/stop a process.

How Processes Start?

- OS uses namespaces to allocate resources (CPU, RAM).
- PID 0: system boot process → init or systemd.
- **For example, once a system boots and it initialises, systemd is one of the first processes that are started. Any program or piece of software that we want to start will start as what's known as a child process of systemd. This means that it is controlled by systemd, but will run as its own process (although sharing the resources from systemd) to make it easier for us to identify and the likes.**
- **systemd starts early and manages child processes.**

Starting, Backgrounding, and Foregrounding processes

Some applications can be started on the boot of the system that we own. For example, web servers, database servers or file transfer servers. This software is often critical and is often told to start during the boot-up of the system by administrators.

Let's use apache2 service as an example

- Managed using systemctl:
 - systemctl start apache2 → start service.
 - systemctl stop apache2 → stop service.
 - systemctl enable / disable → auto-start on boot.
- We can do four options with systemctl:
 - Start
 - Stop
 - Enable
 - Disable

Starting, Backgrounding, and Foregrounding processes

Processes can run in two states: **In the background and in the foreground**. For example, commands that you run in your terminal such as "echo" or things of that sort will run in the foreground of your terminal as it is the only command provided that hasn't been told to run in the background. "Echo" is a great example as the output of echo will return to you in the foreground, but wouldn't in the background - take the screenshot below, for example.

- **Foreground: command runs interactively (e.g., echo "Hi").**
- **Background: add & or use Ctrl+Z to suspend.**
- **Useful for long-running tasks (file copy, scripts).**

Foregrounding Processes

- Use ps aux to confirm background processes.
- Bring process back with **fg**.
- Example:
 - Run script → suspend with Ctrl+Z.
 - Use fg to resume it in the foreground.

My next step

Your Next Step

We have learned a small part of the Linux basics in this course. However, **Linux is much deeper than that.**

What you need to do now is head over to YouTube, **learn more about Linux, use it regularly, and get comfortable with it.**

Linux is a very important part of the cybersecurity field, and you will rely on it heavily.

I recommend that you study the Linux+ course:

<https://youtube.com/playlist?list=PLLLr6jKKdyK1FBi3pLVAmilLvMwWHw-84&si=ROGBGXfdlnSzjG0s>

And also check out the Linux fundamentals courses:

https://youtube.com/playlist?list=PLLLr6jKKdyK15rw72epw7iEDb0DL_af3C&si=HWSlIDfJNuZDJcMR

Windows

I'll eat your ram



Windows 11

What is Windows

Windows..?

Windows is an operating system (OS) developed by Microsoft.

It's one of the most widely used operating systems in the world for personal computers, laptops, and servers.

The first version was released in 1985, and since then, Windows has gone through many versions, becoming more advanced and user-friendly.

It provides a Graphical User Interface (GUI), which makes it easy for users to interact with their computers using windows, icons, and menus.

Why Learn Windows?

- **Most organizations rely on it.**
- **Essential for IT support, system administration, and cybersecurity.**
- **Knowledge of Windows internals is important for pentesting and red teaming**

Windows Editions

One of the most well-known versions was Windows XP, which had a long life cycle and gained wide popularity. After XP, Microsoft released Windows Vista

When Microsoft announced the end-of-life date for XP, many organizations such as corporations and hospitals were forced to migrate to Windows 7, testing it across hardware and devices. This created challenges for vendors and customers, but eventually Windows 7 became one of the most successful versions.

Later came Windows 8 and 8.1, but like Vista, they were short-lived due to design and usability issues.

Then arrived Windows 10, which brought stability, long-term support, and significant improvements.

After that, Windows 11 was introduced on October 5th, 2021, and is currently the latest version for desktop users. Windows 11 comes in two main editions: Home and Pro, each targeting different audiences.

For servers, Microsoft offers the Windows Server line. The latest version is Windows Server 2025, while the VM example we're working on runs Windows Server 2019 Standard.

Windows file system

The file system used in modern versions of Windows is the New Technology File System or simply NTFS

What is NTFS?

NTFS, which stands for **New Technology File System**, is the primary and most modern file system used by Microsoft Windows operating systems. **Think of it as the "rulebook" or the "filing manager" that controls how data is stored, organized, and retrieved on a hard drive, SSD, or USB flash drive.**

NTFS is known as a journaling file system. In case of a failure, the file system can automatically repair the folders/files on disk using information stored in a log file. This function is not possible with FAT.

Key Features and Advantages of NTFS

1. Security and Permissions

2. Support for Large Files and Drives

3. Encryption (EFS)

etc..

Windows file system

Before NTFS, there was FAT16/FAT32 (File Allocation Table) and HPFS (High Performance File System).

- FAT16 – An older file system with limited storage support (up to 2GB).
- FAT32 – Improved version, supports larger partitions (up to 2TB) but has a 4GB maximum file size.
- NTFS (New Technology File System) – Default file system in modern Windows, supports large files, file permissions, encryption, and journaling (better security and reliability).

You can read Microsoft's official documentation on FAT, HPFS, and NTFS [here](#)

NTFS Permissions

On NTFS volumes, you can set permissions that grant or deny access to files and folders.
The permissions are:

- **Full control**
- **Modify**
- **Read & Execute**
- **List folder contents**
- **Read**
- **Write**

Permission	Meaning for Folders	Meaning for Files
Read	Permits viewing and listing of files and subfolders	Permits viewing or accessing of the file's contents
Write	Permits adding of files and subfolders	Permits writing to a file
Read & Execute	Permits viewing and listing of files and subfolders as well as executing of files; inherited by files and folders	Permits viewing and accessing of the file's contents as well as executing of the file
List Folder Contents	Permits viewing and listing of files and subfolders as well as executing of files; inherited by folders only	N/A
Modify	Permits reading and writing of files and subfolders; allows deletion of the folder	Permits reading and writing of the file; allows deletion of the file
Full Control	Permits reading, writing, changing, and deleting of files and subfolders	Permits reading, writing, changing and deleting of the file

Windows \ System32 Folders

In simple terms, the System32 folder is **the core brain and central library of the Windows operating system**. It contains thousands of essential files that Windows needs to function properly.

- **Located in C:\Windows\System32.**
- **Contains essential system files, libraries (DLLs), and executables (EXEs) required for Windows to run.**
- **Includes important tools like cmd.exe, taskmgr.exe, and many services.**
- **If System32 is damaged or deleted, Windows will not function properly.**

Why is it So Important?

- **Stability: Deleting, renaming, or corrupting files in System32 can cause programs to malfunction**
- **Security: Because it's so critical, Windows protects this folder with strong permissions.**
- **Centralization: It provides a single, predictable location for system files, making it easier for Windows and software developers to find the resources they need.**

User Accounts, Profiles, and Permissions

User accounts can be one of two types on a typical local Windows system: Administrator & Standard User. **The user account type will determine what actions the user can perform on that specific Windows system.**

- **An Administrator can make changes to the system: add users, delete users, modify groups, modify settings on the system, etc.**
- **A Standard User can only make changes to folders/files attributed to the user & can't perform system-level changes, such as install programs.**
- **Each person using the computer has a unique account (Local, Microsoft Account, or Domain Account).**
- **Each account has a profile (stored in C:\Users\username) that contains personal files, settings, and preferences.**
- **Windows uses Access Control Lists (ACLs) to define what each user can do (read, write, modify, execute).**

User Access Control (UAC)

User Account Control (UAC) is a security technology that helps prevent unauthorized changes to your Windows operating system. Its primary goal is to stop malware (like viruses, spyware, and ransomware) from making damaging changes to your computer without your knowledge.

Think of it as a "Are you sure?" prompt that appears whenever an application tries to make a change that requires administrative privileges.

- This concept was first introduced with the short-lived Windows Vista and continued with versions of Windows that followed.
- **UAC (by default) doesn't apply for the built-in local administrator account.**

How does UAC work?

When a user with an account type of administrator logs into a system, the current session doesn't run with elevated permissions. When an operation requiring higher-level privileges needs to execute, the user will be prompted to confirm if they permit the operation to run.

Settings and the Control Panel

Control Panel is the traditional, full-featured hub for system administration and advanced settings in Windows. It dates back to the very early versions of Windows.

- Control Panel: Traditional interface to manage system settings (network, hardware, accounts, etc.).

The Settings app was introduced with Windows 8 and greatly expanded in Windows 10 and 11. It is the new, touch-friendly, and user-centric interface for configuring Windows.

- Settings App (introduced in Windows 8/10): Modernized version with a simpler interface.

Both can be used, but Microsoft is slowly moving features from Control Panel into Settings.

Windows tools

System Configuration > utility (MSConfig) is for advanced troubleshooting, and its main purpose is to help diagnose startup issues.

Computer Management > (compmgmt) utility has three primary sections: System Tools, Storage, and Services and Applications.

System information > This tool gathers information about your computer and displays a comprehensive view of your hardware, system components, and software environment, which you can use to diagnose computer issues.

Resource Monitor > Resource Monitor displays per-process and aggregate CPU, memory, disk, and network usage information, in addition to providing details about which processes are using individual file handles and modules

Command prompt > an application (program) that allows users to interact with the operating system by typing commands to perform tasks, run advanced administrative functions, and troubleshoot issues

Windows Registry > is a central hierarchical database used to store information necessary to configure the system for one or more users, applications, and hardware devices.

Windows updates

What is Windows updates?

Windows Update is a service provided by Microsoft to provide security updates, feature enhancements, and patches for the Windows operating system and other Microsoft products, such as Microsoft Defender.

- Updates are typically released on the 2nd Tuesday of each month. This day is called Patch Tuesday. That doesn't necessarily mean that a critical update/patch has to wait for the next Patch Tuesday to be released. If the update is urgent, then Microsoft will push the update via the Windows Update service to the Windows devices.
- Refer to the following link to see the Microsoft Security Update Guide [here](#)

Virus & threat protection

What is Virus & threat protection?

It is a built-in Windows security feature **that helps protect your device against viruses, malware, and other threats. It continuously scans files, applications, and processes running on your system, providing real-time protection. It also allows you to run quick or full system scans, manage quarantined files, and stay updated with the latest security definitions to keep your computer safe.**

Let's try it

Virus & threat protection options

Scan options

- **Quick scan** - Checks folders in your system where threats are commonly found.
- **Full scan** - Checks all files and running programs on your hard disk. This scan could take longer than one hour.
- **Custom scan** - Choose which files and locations you want to check.

Threat history

- **Last scan** - Windows Defender Antivirus automatically scans your device for viruses and other threats to help keep it safe.
- **Quarantined threats** - Quarantined threats have been isolated and prevented from running on your device. They will be periodically removed.
- **Allowed threats** - Allowed threats are items identified as threats, which you allowed to run on your device.

Warning: Allow an item to run that has been identified as a threat only if you are 100% sure of what you are doing.

Virus & threat protection settings

- **Real-time protection** - Locates and stops malware from installing or running on your device.
- **Cloud-delivered protection** - Provides increased and faster protection with access to the latest protection data in the cloud.
- **Automatic sample submission** - Send sample files to Microsoft to help protect you and others from potential threats.
- **Controlled folder access** - Protect files, folders, and memory areas on your device from unauthorized changes by unfriendly applications.
- **Exclusions** - Windows Defender Antivirus won't scan items that you've excluded.
- **Notifications** - Windows Defender Antivirus will send notifications with critical information about the health and security of your device.
- **Warning: Excluded items could contain threats that make your device vulnerable. Only use this option if you are 100% sure of what you are doing.**

Firewall & newtork protection

What is a firewall?

Traffic flows into and out of devices via what we call ports. **A firewall is what controls what is - and more importantly isn't - allowed to pass through those ports.** You can think of it like a security guard standing at the door, checking the ID of everything that tries to enter or exit

Per Microsoft, "Windows Firewall offers three firewall profiles: domain, private and public".

- **Domain** - The domain profile applies to networks where the host system can authenticate to a domain controller.
- **Private** - The private profile is a user-assigned profile and is used to designate private or home networks.
- **Public** - The default profile is the public profile, used to designate public networks such as Wi-Fi hotspots at coffee shops, airports, and other locations.

Bitlocker

What is Bitlocker?

BitLocker is a built-in Windows encryption feature that helps protect your data by encrypting the entire drive. It ensures that even if your computer is lost or stolen, unauthorized users cannot access your files without the proper password, PIN, or recovery key. While you can use BitLocker without extra hardware, it works best when combined with a Trusted Platform Module (TPM) chip.

What is TPM?

A TPM (Trusted Platform Module) is a small security chip built into many modern computers by manufacturers. It's designed to perform security-related tasks, such as storing encryption keys safely and checking the integrity of your system before it starts up.

- **Note: BitLocker provides the most protection when used with a Trusted Platform Module (TPM) version 1.2 or later.**
- **Refer to the official Microsoft documentation to learn more about BitLocker [here](#)**

My next step

Your next step

We have learned a small part of the Windows basics in this course. However, **Windows is much deeper than that.**

What you need to do now is head over to YouTube, **learn more about Windows**

Windows is a very important part of the cybersecurity field (especially for Red Teamers).

I recommend that you this course:

<https://youtube.com/playlist?list=PLsep1uQfemNFvT57-DOjzSnGspJv8efpS&si=q-M5i5HVbfHPh7XI>

Introduction to Cyber Security

CyberSecurity

What is Cybersecurity?

Cybersecurity is the practice of protecting systems, networks, programs, and data from digital attacks. These attacks are usually aimed at accessing, changing, or destroying sensitive information; extorting money from users; or interrupting normal business processes.

Why is Cybersecurity Important?

We live in a digital world. Our daily lives, economies, and critical infrastructure (like power grids and hospitals) rely on connected systems. A successful cyberattack can have severe consequences, including:

- **Financial Loss:** Theft of banking information, cryptocurrency, or funds. Business disruption is also extremely costly.
- **Identity Theft:** Attackers can steal your personal data to impersonate you or sell it on the dark web.
- **Reputation Damage:** For companies, a data breach destroys customer trust.
- **Safety Risks:** Attacks on critical infrastructure (e.g., hacking a power plant or a hospital's systems) can put real lives in danger.

CyberSecurity vs Information Security

While the terms are often used interchangeably, **there is a nuanced difference between them.**

- **Information Security (InfoSec)** is the broader, overarching practice of protecting **all information, whether digital or physical.**
- **Cybersecurity** is a subset of InfoSec that focuses specifically on **protecting digital information in cyberspace from cyberattacks.**

Take a look at this table >

CyberSecurity vs Information Security

Features	InfoSec	Cybersecurity
Primary Focus	Protecting information itself (its confidentiality, integrity, and availability).	Protecting digital systems and networks that store, process, or transmit that information.
Protects Against	All types of threats: unauthorized access, disclosure, disruption, modification, inspection, recording, or destruction.	Primarily digital, external threats: hacking, malware, phishing, DDoS attacks, etc.
Key Concern	Is the data secure? Regardless of how it's stored or who is attacking it.	Is the digital infrastructure secure? Is the network, software, and hardware safe from cyberattacks?

Cybersecurity fields

Cybersecurity is a broad domain with multiple areas of specialization.

- The three main fields are:

1. **Offensive side**
2. **Defensive side**
3. **Administrative side (Out Of Scope)**

- **Offensive** = Finding weaknesses.
- **Defensive** = Protecting systems.
- **Administrative** = Setting rules & strategy.

Offensive security

"To outsmart a hacker, you need to think like one."

What is Offensive Security

Offensive security also known as (Offsec), it involves breaking into computer systems, exploiting software bugs, and finding loopholes in applications to gain unauthorized access. The goal is to understand hacker tactics and enhance our system defences.

Roles include:

- **Penetration Testing**
- **Red Teaming**
- **Ethical Hacking**

Defensive security

What is Defensive security?

Defensive Security is the practice of protecting computer systems, networks, and data from cyberattacks and unauthorized access. It is the proactive and reactive set of measures designed to prevent, detect, and respond to threats.

Think of it as building a fortress:

- **You build walls and gates to keep intruders out (Prevention).**
- **You have guards and watchtowers to spot anyone trying to climb the walls (Detection).**
- **You have a plan and a response team to capture intruders who get inside and repair any damage (Response).**

Security Operations Center (soc)

A Security Operations Center (SOC) is a team of cyber security professionals that monitors the network and its systems to detect malicious cyber security events. Some of the main areas of interest for a SOC are:

- **Vulnerabilities:** Whenever a system vulnerability (weakness) is discovered, it is essential to fix it by installing a proper update or patch. When a fix is unavailable, the necessary measures should be taken to prevent an attacker from exploiting it. Although remediating vulnerabilities is vital to a SOC, it is not necessarily assigned to them.
- **Policy violations:** A security policy is a set of rules required to protect the network and systems. For example, it might be a policy violation if users upload confidential company data to an online storage service.
- **Unauthorized activity:** Consider the case where a user's login name and password are stolen, and the attacker uses them to log into the network. A SOC must detect and block such an event as soon as possible before further damage is done.
- **Network intrusions:** No matter how good your security is, there is always a chance for an intrusion. An intrusion can occur when a user clicks on a malicious link or when an attacker exploits a public server. Either way, when an intrusion occurs, we must detect it as soon as possible to prevent further damage.

Threat Intelligence

Security operations cover various tasks to ensure protection; one such task is threat intelligence.

In this context, intelligence refers to information you gather about actual and potential enemies. A threat is any action that can disrupt or adversely affect a system. Threat intelligence collects information to help the company better prepare against potential adversaries. The purpose would be to achieve a threat-informed defence.

Intelligence needs data. Data has to be collected, processed, and analyzed. Data is collected from local sources such as network logs and public sources such as forums. Data processing arranges it into a format suitable for analysis. The analysis phase seeks to find more information about the attackers and their motives; moreover, it aims to create a list of recommendations and actionable steps.

- **Learning about your adversaries lets you know their tactics, techniques, and procedures. As a result of threat intelligence, we identify the threat actor (adversary) and predict their activity. Consequently, we can mitigate their attacks and prepare a response strategy.**

Digital Forensics

Forensics is the application of science to investigate crimes and establish facts. With the use and spread of digital systems, such as computers and smartphones, a new branch of forensics was born to investigate related crimes: computer forensics, which later evolved into digital forensics.

In defensive security, the focus of digital forensics shifts to analyzing evidence of an attack and its perpetrators and other areas such as intellectual property theft, cyber espionage, and possession of unauthorized content. Consequently, digital forensics will focus on different areas, such as:

- File System: Analyzing a digital forensics image (low-level copy) of a system's storage reveals much information, such as installed programs, created files, partially overwritten files, and deleted files.
- System memory: If the attacker runs their malicious program in memory without saving it to the disk, taking a forensic image (low-level copy) of the system memory is the best way to analyze its contents and learn about the attack.
- System logs: Each client and server computer maintains different log files about what is happening. Log files provide plenty of information about what happened on a system. Even if the attacker tries to clear their traces, some traces will remain.
- Network logs: Logs of the network packets that have traversed a network would help answer more questions about whether an attack is occurring and what it entails.

Incident Response

An incident usually refers to a data breach or cyber attack; however, in some cases, it can be something less critical, such as a misconfiguration, an intrusion attempt, or a policy violation.

Incident response specifies the methodology that should be followed to handle such a case. The aim is to reduce damage and recover in the shortest time possible.

The four major phases of the incident response process are:

1. Preparation: This requires a team trained and ready to handle incidents. Ideally, various measures are put in place to prevent incidents from happening in the first place.
2. Detection and Analysis: The team has the necessary resources to detect any incident; moreover, it is essential to analyze any detected incident further to learn about its severity.
3. Containment, Eradication, and Recovery: Once an incident is detected, it is crucial to stop it from affecting other systems, eliminate it, and recover the affected systems. For instance, when we notice that a system is infected with a computer virus, we would like to stop (contain) the virus from spreading to other systems, clean (eradicate) the virus, and ensure proper system recovery.
4. Post-Incident Activity: After a successful recovery, a report is produced, and the lesson learned is shared to prevent similar future incidents.

Types of Malware

Malware stands for malicious software. Software refers to programs, documents, and files you can save on a disk or send over the network. Malware includes many types, such as:

- A virus is a piece of code (part of a program) that attaches itself to a program. It is designed to spread from one computer to another and works by altering, overwriting, and deleting files once it infects a computer. The result ranges from the computer becoming slow to unusable.
- Trojan Horse is a program that shows one desirable function but hides a malicious function underneath. For example, a victim might download a video player from a shady website that gives the attacker complete control over their system.
- Ransomware is a malicious program that encrypts the user's files. Encryption makes the files unreadable without knowing the encryption password. The attacker offers the user the encryption password if the user is willing to pay a "ransom."

Malware Analysis

What is Malware Analysis?

Malware analysis is the process of studying malicious software to understand:

- How it works
- What it is designed to do
- How to detect, contain, and remove it

It's a critical skill in cybersecurity, especially in incident response, forensics, and threat intelligence.

Malware analysis aims to learn about such malicious programs using various means:

- 1. Static analysis works by inspecting the malicious program without running it. This usually requires solid knowledge of assembly language (the processor's instruction set, i.e., the computer's fundamental instructions).**
- 2. Dynamic analysis works by running the malware in a controlled environment and monitoring its activities. It lets you observe how the malware behaves when running.**

Careers in Cybersecurity

Cyber security careers are becoming more in demand. There are many different jobs within the security industry, from offensive security (hacking machines and reporting on vulnerabilities) to defensive security (defending against and investigating cyberattacks).

Why Cybersecurity career?

- **Cybersecurity is one of the fastest-growing fields.**
- **Wide range of career paths available.**
- **Roles cover technical, analytical, and managerial areas.**

Security Analyst

Responsible for maintaining the security of an organisation's data

Security Analysts **are the frontline defenders and digital sentinels of an organization.** They are integral to constructing security measures across organisations to protect the company from attacks. **Analysts explore and evaluate company networks to uncover actionable data and recommendations for engineers to develop preventative measures.** This job role requires working with various stakeholders to gain an understanding of security requirements and the security landscape.

Responsibilities

- Working with various stakeholders to analyse the cyber security throughout the company
- Compile ongoing reports about the safety of networks, documenting security issues and measures taken in response
- Develop security plans, incorporating research on new attack tools and trends, and measures needed across teams to maintain data security.

[Becoming a Cyber Security Analyst](#)

Security Engineer

Design, monitor and maintain security controls, networks, and systems to help prevent cyberattacks

Security engineers **develop and implement security solutions using threats and vulnerability data - often sourced from members of the security workforce.** Security engineers work across circumventing a breadth of attacks, including web application attacks, network threats, and evolving trends and tactics. The ultimate goal is to retain and adopt security measures to mitigate the risk of attack and data loss.

Responsibilities

- **Testing and screening security measures across software**
- **Monitor networks and reports to update systems and mitigate vulnerabilities**
- **Identify and implement systems needed for optimal security**

[How to Become a Security Engineer](#)

Incident Responder

Identifies and mitigates attacks whilst an attacker's operations are still unfolding

An incident usually refers to a data breach or cyber attack; however, in some cases, it can be something less critical, such as a misconfiguration, an intrusion attempt, or a policy violation.

Incident responders respond productively and efficiently to security breaches. Responsibilities include creating plans, policies, and protocols for organisations to enact during and following incidents

Responsibilities

- **Developing and adopting a thorough, actionable incident response plan**
- **Maintaining strong security best practices and supporting incident response measures**
- **Post-incident reporting and preparation for future attacks, considering learnings and adaptations to take from incidents**

Digital Forensics

Responsible for using digital forensics to investigate incidents and crimes

If you like to play detective, this might be the perfect job. If you are working as part of a law-enforcement department, you would be focused on collecting and analysing evidence to help solve crimes: charging the guilty and exonerating the innocent. On the other hand, if your work falls under defending a company's network, you will be using your forensic skills to analyse incidents, such as policy violations.

Responsibilities

- Collect digital evidence while observing legal procedures
- Analyse digital evidence to find answers related to the case
- Document your findings and report on the case

Malware Analyst

Analyses all types of malware to learn more about how they work and what they do

A malware analyst's work involves analyzing suspicious programs, discovering what they do and writing reports about their findings. A malware analyst is sometimes called a reverse-engineer as their core task revolves around converting compiled programs from machine language to readable code, usually in a low-level language. This work requires the malware analyst to have a strong programming background, especially in low-level languages such as assembly language and C language. The ultimate goal is to learn about all the activities that a malicious program carries out, find out how to detect it and report it.

Responsibilities

- Carry out static analysis of malicious programs, which entails reverse-engineering
- Conduct dynamic analysis of malware samples by observing their activities in a controlled environment
- Document and report all the findings

Penetration Tester

Responsible for testing technology products for security loopholes

You may see penetration testing referred to as pentesting and ethical hacking. A penetration tester's job role is to test the security of the systems and software within a company - this is achieved through attempts to uncover flaws and vulnerabilities through systemised hacking. Penetration testers exploit these vulnerabilities to evaluate the risk in each instance. The company can then take these insights to rectify issues to prevent a real-world cyberattack.

Responsibilities

- **Conduct tests on computer systems, networks, and web-based applications**
- **Perform security assessments, audits, and analyse policies**
- **Evaluate and report on insights, recommending actions for attack prevention**

[How to Become a Penetration Tester](#)

Red Teamer

Plays the role of an adversary, attacking an organisation and providing feedback from an enemies perspective

Red teamers share similarities to penetration testers, with a more targeted job role. **Penetration testers look to uncover many vulnerabilities across systems to keep cyber-defence in good standing, whilst red teamers are enacted to test the company's detection and response capabilities. This job role requires imitating cyber criminals' actions, emulating malicious attacks, retaining access, and avoiding detection. Red team assessments can run for up to a month, typically by a team external to the company. They are often best suited to organisations with mature security programs in place.**

Responsibilities

- Emulate the role of a threat actor to uncover exploitable vulnerabilities, maintain access and avoid detection
- Assess organisations' security controls, threat intelligence, and incident response procedures
- Evaluate and report on insights, with actionable data for companies to avoid real-world instances

[Red Teaming](#)

My next step

Your next step

After learning about what cybersecurity is, its relationship with information security, and getting a brief introduction to the offensive and defensive sides and their fields, **this alone is not enough to consider yourself as having a foundation in information security!**

You need to dive deeper into the fundamentals of information security and get to know them up close. I recommend the CompTIA Security+ course, as it can provide you with exactly what you need.

I recommend CompTIA Security+ 701 from Netriders Academy:

<https://netriders.academy/all-courses/security-new/>

- **You should go for CompTIA Security+ before diving into cybersecurity fields**
- **HR love seeing CompTIA Security+ on resumes**

So What's next..

What's Next After the Fundamentals?

Alright, so far we've learned computer fundamentals, networking, operating systems (Windows & Linux), and studied information security basics. So, **what comes after all this?**

The answer might not be what you think. **It's not a linear path where you complete one stage and simply ask, "What's next?"** The key lies in your ability to make a decision and find the field that best suits you.

This is where I'll explain how you can determine your specialization in cybersecurity.

The Simple Answer: Experimentation

At this stage, you should know that several platforms allow you to try different cybersecurity fields hands-on. A prime example is the platform we're using now: TryHackMe.

I've previously shared various learning paths with you for different fields like SOC Analyst, Penetration Testing, and Red Teaming. **My advice is to dive in, explore, and discover what genuinely interests you. See which area you naturally gravitate towards.**

Once you've explored and feel drawn to a specific area -whether offensive or defensive- **I highly recommend studying an introduction to penetration testing.**

The most renowned certification that provides this foundation is the eJPTv2 (eLearnSecurity Junior Penetration Tester v2). It's a certificate that covers multiple fascinating topics in penetration testing, and I see it as an ideal starting point, a crucial first step that logically leads to more advanced ones.

eJPTv2..

You can prepare for this certification through:

- [INE \(The official provider\)](#)
- [Netriders Academy](#)

And this is a file that you should go for (Created by Shatha Alosaimi)

<https://x.com/PTShatha511/status/1900238869062205457>

Pro Tip: If you just want to purchase the exam voucher alone, you can get it directly to this [link](#) for getting eJPTv2 with only **\$116**. Also, make sure to **follow [INE's official X account](#)** for promotions and discounts, so you can buy the voucher at the lowest possible price.

A roadmap?

This is the roadmap that I found most suitable and have personally followed. You can certainly follow any roadmap that appeals to you, but remember that **networking, operating systems, information security fundamentals, and programming are extremely important regardless of which specific roadmap you choose.**

You can access roadmap from this link:

https://drive.google.com/file/d/11M_dMQmXdGSNVlhb0V8sEOUkhMtOGqIC/view?usp=drivesdk

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- **Linux Fundamentals Part 2:** <https://tryhackme.com/room/networkingcoreprotocols>
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- **Offensive Security Intro:** <https://tryhackme.com/room/offensivesecurityintro>
- **Defensive Security Intro:** <https://tryhackme.com/room/defensivesecurityintro>
- **Careers in CyberSecurity:** <https://tryhackme.com/room/careersincyber>
- **Security+ course:** <https://netriders.academy/all-courses/security-new/>
- **eJPTv2 course (INE):** [course](#)
- **eJPTv2 course (Netriders academy):** [course](#)
- **eJPTv2 tips file:** <https://x.com/PTShatha511/status/1900238869062205457>

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Thank You

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